CHECK WHAT YOU LEARNT

Check Yourself - Unit 1

A. Choose the correct answer

1. A signal that can only be ON or OFF is called
   (a) analog (b) digital (c) electric (d) linear
2. Room temperature is best described as
   (a) digital (b) analog (c) neither (d) both
3. A resistor banded red - red - brown has a value of
   (a) 22 ohm (b) 220 ohm (c) 2200 ohm (d) 10K
4. The VCC pin of a sensor module connects to
   (a) GND (b) 5V (c) A0 (d) D13
5. The AO pin of a module gives
   (a) only 0 or 1 (b) a number from 0 to 1023 (c) 5 volts always (d) nothing
6. The small blue screw on a module adjusts
   (a) the AO reading (b) the DO threshold (c) the power (d) the colour

B. Fill in the blanks

1 Analog pins on the Arduino Uno are labelled ______ to ______.

2 The third colour band on a resistor tells you how many ______________ to add.

3 Brown - black - orange is a ______________ ohm resistor.

4 Every sensor module must always have ______________ and GND connected.

C. Answer in one or two lines

1 Give two examples of analog quantities in your own home, and say why they are analog.

2 Why can a machine that understands only digital signals never describe rain properly?

3 A friend says the AO reading changed after turning the blue screw. Are they right? Explain.

4 Write down what red - red - brown means, digit by digit.

UNIT 2

The Arduino IDE

Sessions 3, 4 and 5

You already know how to make a pin go on and off. This unit teaches the three skills that separate Class 7 from Class 6: reading a number from a sensor, printing it so you can see it, and turning it into a smooth output.

By the end of this unit you will be able to:

- Upload a program and use the Serial Monitor to watch live values.
- Explain what analogRead() returns and why the range is 0 to 1023.
- Use map() to convert one range of numbers into another.
- Use analogWrite() and PWM to fade an output instead of switching it.

SESSION 3

one class

The Board, the Software and the First Upload

Today you will

- Refresh the parts of the Arduino Uno, especially the analog pins.
- Set the board and port correctly in the IDE.
- Upload Blink and confirm your whole setup works.

The Arduino Uno is a small computer with no screen and no keyboard. It runs exactly one program at a time, keeps it in memory when the power is removed, and starts running again the instant power returns. This year, pay particular attention to two groups of pins: the analog inputs along the bottom, and the digital pins marked with a small **~** symbol.

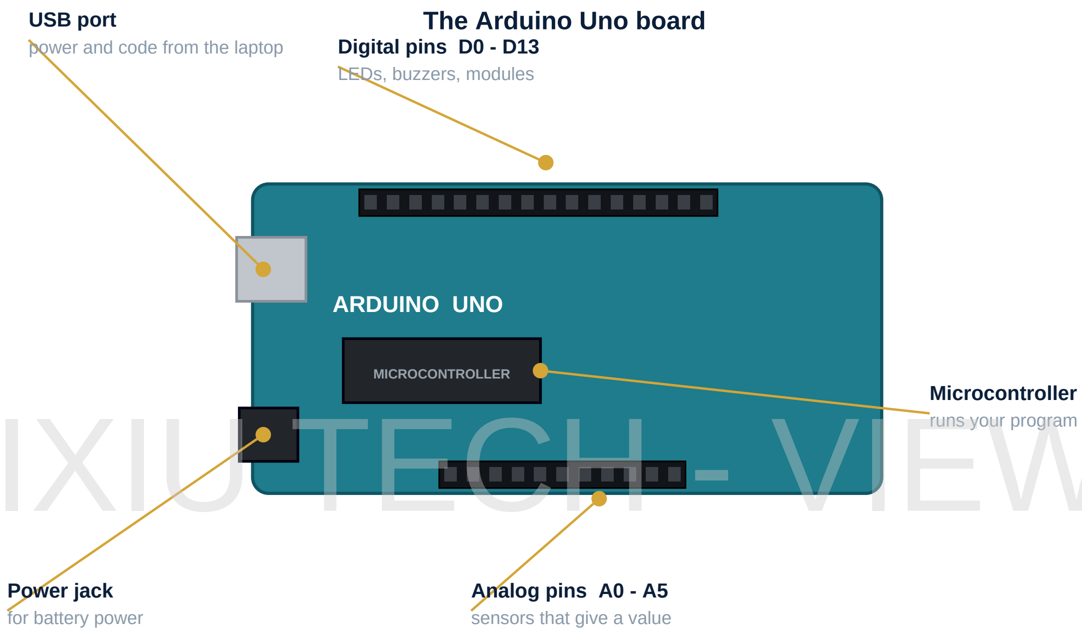


Figure 2.1 - Know your board before you wire anything to it

Group of pins	What is special about them	Used in this book for
A0 to A5	They measure a voltage and report 0 to 1023	the potentiometer, the LDR, the rain sensor
~3, ~5, ~6, ~9, ~10, ~11	They can fade an output, not just switch it	the LED dimmer and mood light
D2 to D13 (all)	Plain on-or-off outputs and inputs	LEDs, the buzzer, the DHT11
5V and GND	Power out to every module you connect	everything, always

Figure 2.2 - Verify checks your code; Upload sends it to the board

> **Watch Out**
> Before your first upload of the year, check **Tools > Board** says Arduino Uno and **Tools > Port** has a port selected. When an upload fails, this is the first thing to check, every single time.

Upload the Blink sketch from Class 6 as a test. If pin 13 blinks, your board, your cable, your laptop and your software are all working. Never start a new project without knowing that.

SESSION 4

one class

Variables and the Serial Monitor

Today you will

- Learn what a variable is and why int and float are different.
- Open the Serial Monitor and watch live numbers.
- Read your first real sensor value from the potentiometer.

A **variable** is a named box that holds a number. You have already used them: `int led = 13;` makes a box called led and puts 13 in it. This year your boxes will hold values that keep changing, because they come from sensors.

You write	It holds	Example
int	A whole number, with no decimal point	int light = 743;
float	A number with a decimal point	float temp = 27.4;
boolean	Only true or false	boolean raining = false;

Choose the right type. A temperature of 27.4 degrees stored in an `int` becomes 27, and the .4 is lost forever. That does not matter much for a classroom thermometer, but it would matter a great deal in a hospital.

4.1 Talking to the Laptop

Until now your machine has had no way of telling you what it is thinking. The Serial Monitor fixes that. Two instructions are all you need: `Serial.begin(9600);` inside setup(), which opens the conversation, and `Serial.println(value);` inside loop(), which sends one number and starts a new line.

Serial Monitor 9600 baud

```
Temperature: 27.0 C   Humidity: 58 %
Temperature: 27.0 C   Humidity: 58 %
Temperature: 27.2 C   Humidity: 59 %
Temperature: 28.4 C   Humidity: 61 %
Temperature: 31.6 C   Humidity: 63 %
Temperature: 34.1 C   Humidity: 64 %  <- hand on sensor
Temperature: 33.0 C   Humidity: 62 %
```

Autoscroll on | Tools > Serial Monitor

Figure 2.3 - The Serial Monitor is your window into the machine's mind

ReadTheKnob.ino

```
int potPin = A0;

void setup() {
  Serial.begin(9600);        // open the line to the laptop
}

void loop() {
  int value = analogRead(potPin);   // 0 to 1023
  Serial.print("Knob reads: ");
  Serial.println(value);            // println starts a new line
  delay(200);
}
```


Try This

Wire the potentiometer: left leg to 5V, right leg to GND, middle leg to A0. Upload the program above and open **Tools > Serial Monitor**. Turn the knob slowly from one end to the other and watch the numbers. Write down the lowest and the highest number you see.

Watch Out

The number at the bottom right of the Serial Monitor window must say **9600**. If it says anything else you will see meaningless symbols instead of numbers. The number in your code and the number in the window have to match.

SESSION 5

one class

analogRead, map and PWM

Today you will

- Understand why analogRead gives 0 to 1023 and analogWrite needs 0 to 255.
- Use map() to convert between the two ranges.
- Fade an LED smoothly using PWM.

Here is a puzzle that catches almost everybody. `analogRead()` gives you a number from **0 to 1023**. But `analogWrite()` only accepts a number from **0 to 255**. Feed one into the other and your LED reaches full brightness after a quarter of a turn and then does nothing for the rest.

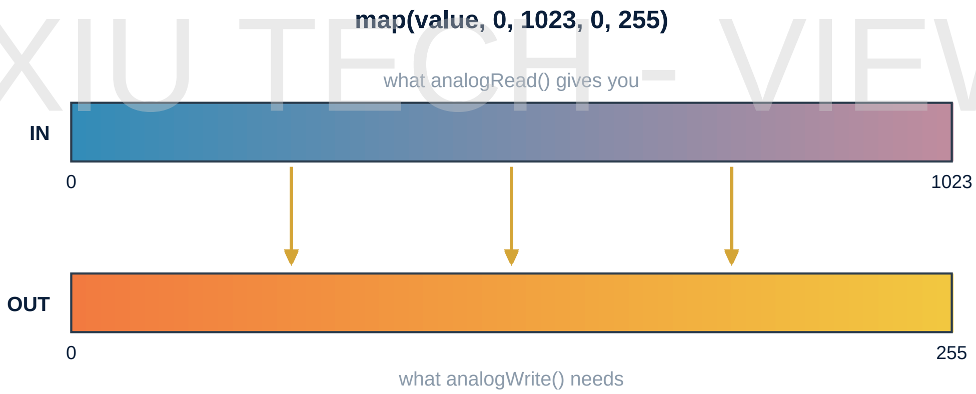


Figure 2.4 - map() stretches one range neatly onto another

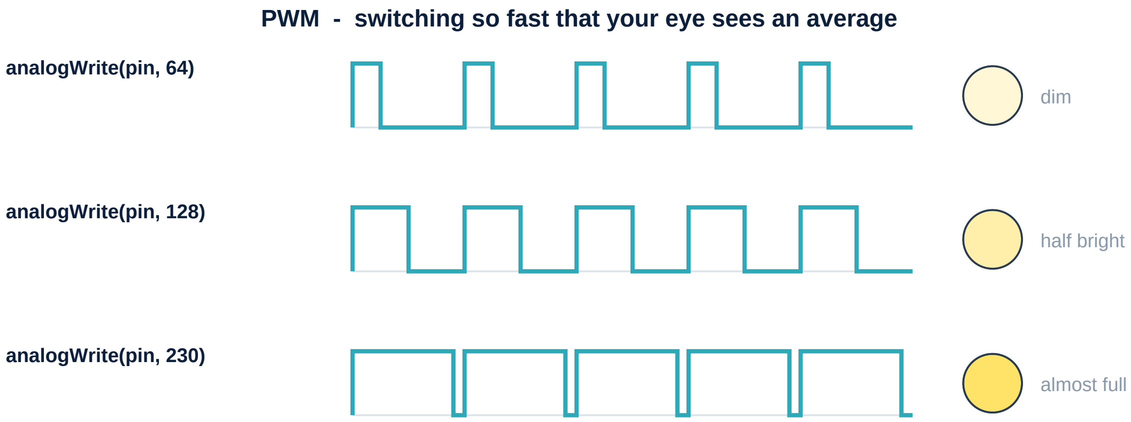
The fix is a single instruction: `map()`. You tell it the value, the range it came from, and the range you want it in. It does the arithmetic for you.

The three lines that matter

```
int raw = analogRead(A0);              // 0 to 1023
int bright = map(raw, 0, 1023, 0, 255); // now 0 to 255
analogWrite(9, bright);                 // pin 9 has a ~ symbol
```

5.1 What PWM Really Does

A digital pin cannot produce half of 5 volts. It can only be fully on or fully off. So how does an LED get dimmer? The board cheats, brilliantly. It switches the pin on and off hundreds of times a second. If the pin is on for a quarter of the time, the LED receives a quarter of the energy, and your eye - which is far too slow to see the switching - simply reports 'dim'.



The pin is still only ON or OFF. It is the amount of time spent ON that changes.
Only pins marked with ~ can do this: D3, D5, D6, D9, D10 and D11.

This trick is called **Pulse Width Modulation**, or PWM, and it is everywhere: in the brightness of your phone screen, in the speed of a drone motor, in the dimmer of an LED bulb. Only six pins on the Uno can do it, and they are all marked with a small ~ next to the number.

Did You Know

PWM is why a cheap LED bulb sometimes looks like it is flickering in a video, even though it looks steady to your eye. The camera's shutter catches the off moments; your eye blurs them together.

New word	What it means
Variable	A named box that holds a number.
int / float	A whole number / a number with a decimal point.
Serial Monitor	The window where a machine prints what it is thinking.
map()	An instruction that stretches one range of numbers onto another.
PWM	Switching a pin on and off very fast so the output appears in-between.